DIGITAL BATTERY PASSPORT

DBP-NASA-B0018

Battery B0018 · Status repurposed · Second-life grade A

VERIFIED: all 14 events pass the six checks

Access tier: legitimate interest (full). Schema aligned with Regulation (EU) 2023/1542, Article 77 and Annex XIII.

Scan to open the live record and check that it matches this copy.

General information

model	B0018
manufacturer	Unknown (NASA Ames test cells)
manufacture date	2008-06-07
chemistry	Li-ion (LiCoO2-based 18650)
form factor	18650 cylindrical
rated capacity ah	2
nominal voltage v	3.7
manufacturing site	not disclosed (laboratory test cell)

Carbon footprint

kgco2e per kwh	75
total kgco2e	0.555
basis	assumed mid value of published 61-106 kg CO2e/kWh (IVL)

Performance and durability

rated capacity ah	2
initial capacity ah	1.84
initial resistance ohm	0.152
expected cycle life	not recorded
cut off voltage v	not recorded

Battery status

status	repurposed
events	14
last event	SOH_ASSESSMENT

State of health (restricted)

assessment: cycle	119
assessment: soh	0.6926
assessment: soh lo	0.631
assessment: soh hi	0.7542
assessment: soh measured	0.6795
assessment: resistance growth	1.045
assessment: method	Gradient Boosting on charge-phase signals, 95% conformal interval
rul: cycle	40
rul: threshold	0.8
rul: rul cycles	7.1
rul: rul lo	0

rul: rul hi	18.8
rul: method	quantile gradient boosting, conformalised (80% band)

Lifecycle decision (restricted)

cycle	59
grade	A
application	EV re-use / light e-mobility
confidence	low
recommendation	Likely EV re-use / light e-mobility, but uncertainty spans grade B: run a laboratory capacity test first
service years	not recorded
co2 avoided kg per 60kwh pack	3606

Signed event chain (14 events)

#	Date	Event	Signed by	Hash
0	2008-06-07	Manufactured	MFR-01	`c08d69952ba1fd78e689…`
1	2008-06-08	Carbon Footprint Declared	MFR-01	`911a8b26bbecaaf39236…`
2	2008-07-07	First Life Start	OEM-01	`0198c6ce35bcb29dc0c3…`
3	2008-07-12	Soh Assessment	AI-ASSESSOR	`ca8a7624637c5893b26d…`
4	2008-07-12	Rul Prediction	AI-ASSESSOR	`9f304cdedcef1664e85c…`
5	2008-07-18	Soh Assessment	AI-ASSESSOR	`d166ba5c993d277f7ba4…`
6	2008-07-18	Rul Prediction	AI-ASSESSOR	`fe4c07c162409adb8fe5…`
7	2008-08-02	First Life Retired	OEM-01	`817cebf95e1b94131c48…`
8	2008-08-02	Soh Assessment	AI-ASSESSOR	`284717b6e1301cd1dd8d…`
9	2008-08-02	Second Life Grading	AI-ASSESSOR	`bb51288c0e575b19c17d…`
10	2008-08-02	Repurposed	REP-01	`002f48f2df46dbef2eb9…`
11	2008-08-06	Soh Assessment	AI-ASSESSOR	`c422f0ce9f51d3048510…`
12	2008-08-10	Soh Assessment	AI-ASSESSOR	`a010cfe9af12802583b2…`
13	2008-08-16	Soh Assessment	AI-ASSESSOR	`5a1742bd630a87f5019c…`

Head hash (fingerprint of the whole chain)
`5a1742bd630a87f5019c983440f6174dbaab8a5b091b91c46e711b24f3d915a6`

Each event stores the SHA-256 hash of the event before it and an Ed25519 signature from the actor who issued it. Changing, deleting or re-ordering any event changes the head hash and fails verification. This printed copy is a snapshot; the live record is authoritative.